HILAL MUHAMAD ABDUL GANI

Bandung, Jawa Barat | 081563955598 | hilalabdulgani@gmail.com
linkedin.com/in/hilal-muhamad | https://hilalmuhamad.my.id/

SUMMARY

Informatics Engineering graduate with hands-on web development and network infrastructure experience across industry roles, government, and independent projects. Proficient in HTML, CSS, JavaScript, PHP, Laravel, FastAPI, React, and Flutter, with strong database skills in MySQL, PostgreSQL, and SQL Server. Experienced in mobile application architecture, enterprise-scale data system design, and network performance optimization. Delivered measurable impact across diverse technical domains by successfully architecting production-ready systems. A rapid learner, adaptable across new tools, and committed to delivering scalable software solutions in fast-paced environments.

WORK EXPERIENCE

BLSDM Komdigi Bandung
March 2026 – June 2026

IT Support Intern

- Designed and deployed a multi-floor network infrastructure using enterprise-grade Ruijie and Linksys hardware to support facility expansion.
- Executed physical network installations, including precise RJ45 cable crimping (Cat5e/Cat6) and structured cabling management to ensure reliable wired connectivity for devices.
- Diagnosed and resolved signal interference by performing systematic wireless site surveys and network mapping with NetSpot and WiFi Analyzer.
- Improved wireless connection stability and throughput via intensive packet-loss and latency testing using PingPlotter.
- Collaborated with IT engineers to design and simulate VLAN segmentation in Packet Tracer and GNS3, ensuring zero-defect production rollouts.

Maqdis Academy
June 2026 – September 2026

Software Engineer Intern

- Developed Flutter mobile interfaces for course directories, Tajwid search, and an audio player with background execution support.
- Built REST API integrations for student registration flows, audio recording uploads, and data synchronization.
- Optimized app performance by implementing local caching for audio metadata and user progress history to support offline access.
- Executed rigorous API and UI testing via Postman to verify push notifications, notification scheduling, and data integrity.

I Channel TV Bandung
April 2022 – June 2022

Assistant Producer Intern

- Supported end-to-end television production workflows, maintaining structured schedules and strict quality control.
- Coordinated effectively with diverse cross-functional teams to execute broadcast projects and meet tight industry deadlines.

PROJECTS

Tracer Study Data Warehouse & OLAP Analytics System
Fullstack Developer

https://github.com/hilalmuhamad/tugas_akhir.git

- Designed and implemented a Star Schema Data Warehouse comprising centralized fact tables and conformed dimensions, integrated via an automated ETL (Extract, Transform, Load) pipeline that maintained relational integrity with zero orphan records.
- Implemented a Hybrid Transactional/Analytical Processing (HTAP) environment using PostgreSQL, Laravel and React js, successfully isolating real-time operational data (OLTP) from the analytical data warehouse to ensure zero performance degradation during complex aggregations.

- Built OLAP dashboards with multi-dimensional querying capabilities to visualize alumni employability metrics and track institutional KPIs.
- Optimized database scalability and query performance through strategic schema separation (public, staging, dwh), table partitioning by cohort year, and the implementation of Block Range Indexes (BRIN) on temporal dimensions.

Focus Talk – Android-Based Smart Intervention & Micro-Learning Nov 2025 – Feb 2026
Mobile Developer
https://github.com/rdwnsyh/focustalk_app.git

- Built a full-stack Android intervention system using Flutter (client) and FastAPI/Python (backend), designed to reduce digital procrastination among students through real-time behavioral intervention.
- Engineered a Heuristic Rule-Based detection engine using Android's UsageStatsManager to monitor foreground app activity in real time and accurately classify distraction-triggering applications (e.g., Instagram, TikTok).
- Developed a Background Service leveraging System Alert Window (WindowManager) to overlay non-blocking English micro-learning quizzes, requiring quiz completion as a soft-gate before accessing distraction apps.
- Designed a FastAPI backend to synchronize user scores, profiles, and a leaderboard system, adding a gamification layer that incentivizes consistent engagement with the learning content.

Maggot Cycle – Distributed Backend for Organic Waste Reporting System Jan 2026
Backend/Infra Engineer

- Architected a distributed microservices backend using Golang (Fiber) and Podman containerization, implementing horizontal scaling with dual backend instances behind an Nginx Load Balancer to handle high-concurrency reporting traffic from hundreds of B2B partner kitchens under a national food-distribution program.
- Designed a PostgreSQL Master-Slave replication architecture and disaster recovery strategy, achieving near-zero data loss (RPO) and sub-1-hour recovery time (RTO), validated through concurrent load testing that maintained a 0% error rate.
- Implemented multi-layer security hardening JWT-based API authentication, Cloudflare WAF, Suricata IDS/IPS, Fail2Ban, and Tailscale VPN for private server access — verified through Nmap-based penetration testing confirming zero exposed internal ports.

EDUCATION

Universitas Logistik dan Bisnis Internasional (Poltekpos) *Bandung, Indonesia | 2023 – 2026*
Diploma (D3) in Informatics Engineering | GPA: 3.72/4.00

- Participated in a Community Service Program (PKM) at SMK Negeri 2 Cimahi focused on improving web programming skills through a Laravel bootcamp.
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming (OOP), Web & Mobile Development, Database Management Systems (SQL/NoSQL), Software Engineering.

TRAININGS AND CERTIFICATIONS

Coursera | *Foundations of Cybersecurity* Jul 2026
SAP | *Introduction to the ABAP Workbench* Jul 2025
Dicoding | *Back-End Development with JavaScript* Sep 2024
Dicoding | *Cloud Practitioner Essentials* Jun 2024

ORGANIZATIONAL EXPERIENCE

Industrial Visit Committee | PT Computrade Teknologi Internasional Aug 2025
Project Officer

- Directed a committee of 24 members to organize a strategic industrial visit, serving as the primary liaison between the university and corporate representatives.
- Managed the event timeline, logistics, and correspondence, ensuring a seamless experience for 76 student participants.

HIMATIF (Informatics Engineering Student Association) March 2025 - Present

Staff Public Relations

- Delivered an internal training session on sponsorship prospecting strategies to 10–30 HIMATIF committee members, sharing practical frameworks for identifying and approaching potential corporate sponsors for student-led events.
- Initiated and managed the "Safari Himpunan" (Comparative Study programs) with external university student associations.
- Acted as the primary liaison between the student association and corporate sponsors, actively managing professional communications and negotiating strategic partnerships to support student-led events.

K-Radio ULBI March 2025 - Present

Head of Media Design Division

- Led a creative team in producing digital content, resulting in a consistent visual identity across all social media platforms.
- Oversaw the design workflow for campus event promotions, increasing audience engagement through visually compelling graphics.

LANGUAGES & SKILLS

- Programming Languages: JavaScript, Python, PHP, Dart, SQL
- Frontend & Mobile: Flutter (BLoC, Clean Architecture), React.js
- Backend Development: Node.js (Express), Laravel, FastAPI
- Databases: PostgreSQL, MySQL, MongoDB, SQLite
- DevSecOps & Infrastructure: Podman, Nginx, Cloudflare WAF, Suricata IDS/IPS, Tailscale, VLAN, System Hardening, Load Balancing
- Tools & Methodologies: Git, Postman, Figma, VS Code
- Languages : Indonesian (Native), English (Intermediate)